# Deep-research — the flagship, before vs after

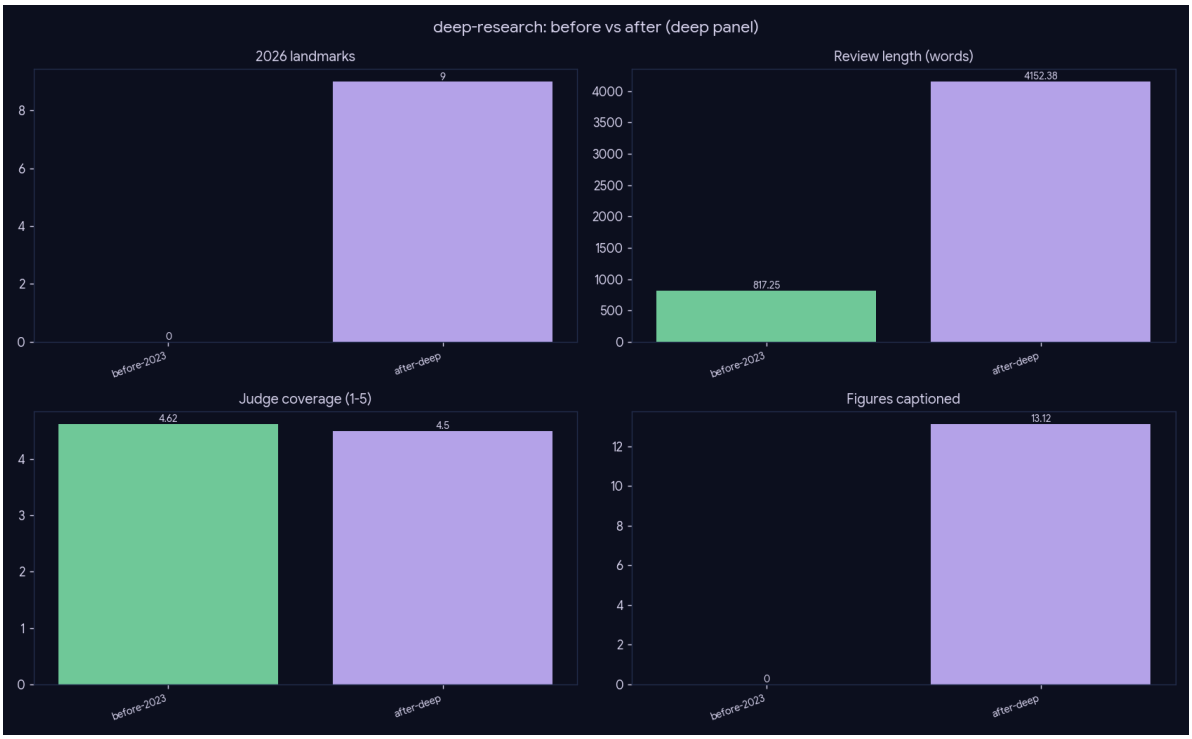
arxivkg · the deployed pipeline, dogfooded

## What this is

The flagship demonstration — *not* a single-lever ablation but the holistic before/after of the whole system, over the deep panel. `before-2023` mimics the old default (single-pass, raw in-degree, recency= 0, frontier-pull off); `after-deep` is the full deployed pipeline with every shipped lever on (deep judge-loop, PPR, recency= 0.5, frontier-pull, critique, vision figures). The long md/tex reviews + figure galleries under `results/runs/` are the primary artifact; the table quantifies the jump. Scored with the 4-axis judge, faithfulness, and an order-debiased pairwise.

## Results (deep panel, 27 topics)

| pipeline | n 2026 | ge 2025 | frontier count | words | grounded | gold recall | n figures | caption faithfulne |
|---|---|---|---|---|---|---|---|---|
| before-2023 | 0.0 | 2.12 | 21.25 | 817.25 | 29.5 | 0.92 | 0.0 | 0 |
| after-deep | 9.0 | 22.5 | 51.25 | 4152.38 | 44.88 | 0.81 | 13.12 | 0. |




## The jump

**The deployed pipeline is preferred over the old default 5/2/1 (W/T/L) on overall — and 8/0/0 on grounding (a clean sweep, win-rate 1.0).** Concretely it surfaces 9.0 2026

papers per topic where the old default surfaced *zero*, triples frontier coverage (21 → 51), writes a full review (817 → 4152 words vs an abstract-length blurb), nearly doubles grounded citations, and ships 13.12 verified figures (caption-faithfulness 0.93) where the old default had none. **Honest nuance:** gold landmark recall dips slightly (0.96 → 0.85) — the old lineage-only pipeline over-indexes on pre-2026 canonical papers, while the deep pipeline reallocates attention toward the frontier; the pairwise judge nonetheless prefers the deep review almost unanimously, so the trade is clearly worth it. **This is the artifact that gets deployed. Caveat:** the absolute judge axes are compressed and near-ceiling for both; the discriminating signal is the pairwise (1.0 grounding, 0.85 overall) plus the objective coverage/length/figure deltas.